

ACUTE APPENDICITIS & INTESTINAL OBSTRUCTION

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ACUTE APPENDICITIS

Anatomy

- The appendix is a worm-shaped blind-ending tube that arises from the posteromedial wall of the caecum 2 cm below the ileocaecal valve.
- It varies in length from 6–9 cm.

McBurney's point:

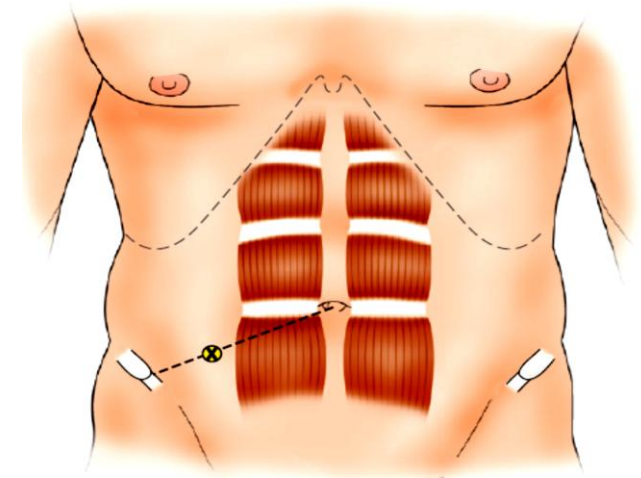
On the surface of the abdomen, this point lies **one-third** of the way along a line drawn between the right anterior superior iliac spine and the umbilicus

BLOOD SUPPLY:

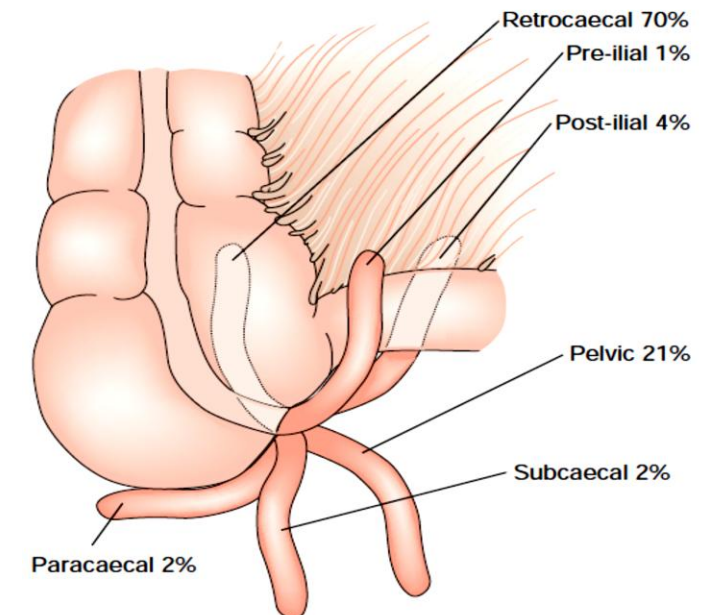
The appendix has its own mesentery, the mesoappendix, and its blood supply comes from *the appendicular artery*.

Position:

The most common site is **retrocaecal**, but data from diagnostic laparoscopy indicate that the pelvic position is probably common.



McBurney's point.



Variations in the position of the appendix.

Aetiology

Several different mechanisms have been proposed:

Diet:

One of the more popular causes being a diet ;lacking in fibre and a consequent slow transit time and alteration in bacterial flora.

Viral infection:

It may be an aetiological agent, as there is an association between appendicitis and concurrent viral illness and because there is a seasonal variation in the incidence of appendicitis.

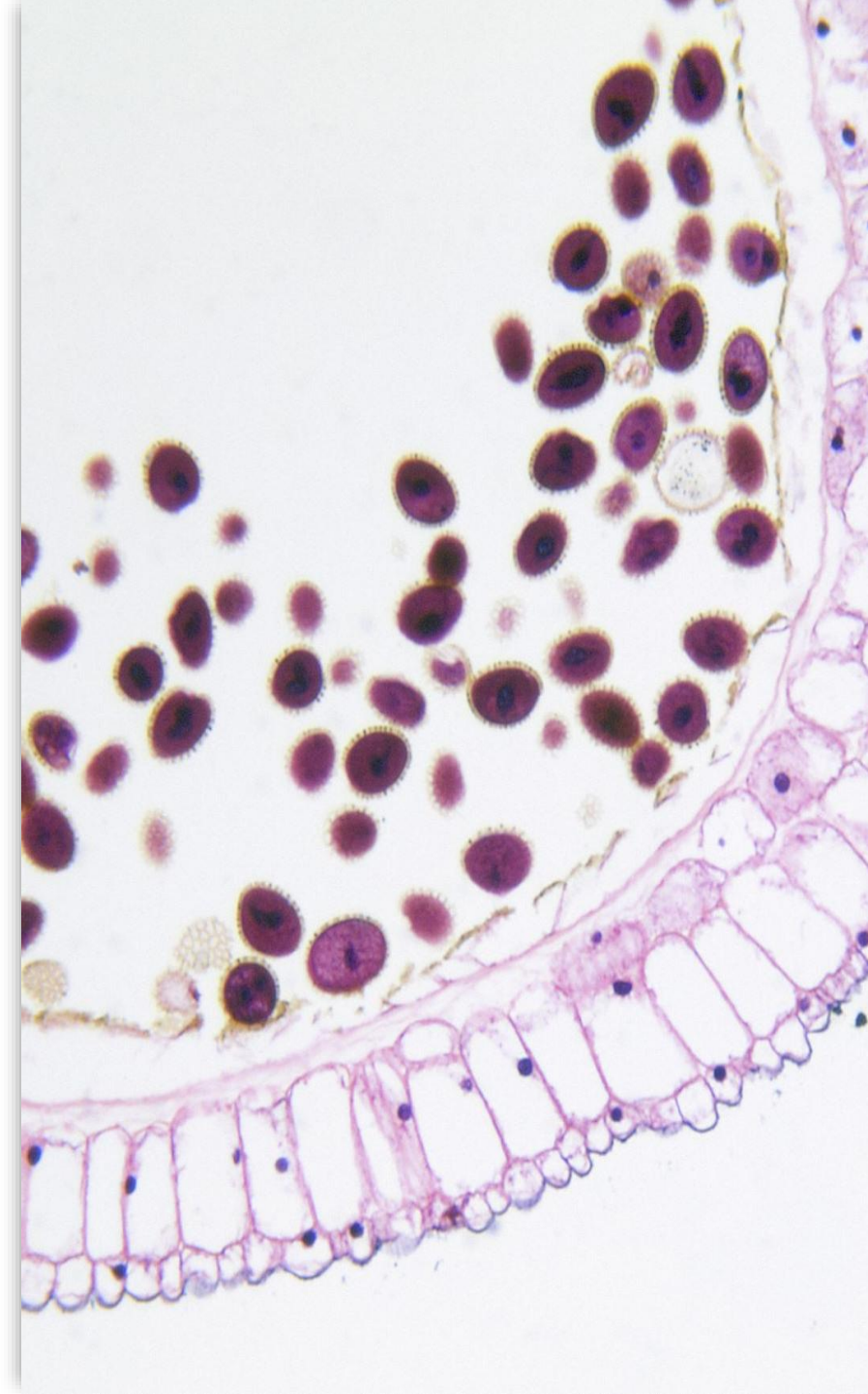
Pathogenesis

Faecoliths, lymphoid hyperplasia, foreign bodies, carcinoid tumours  **Appendicular Lumen Obstruction**  Wall is inflamed  *Fibrinopurulent exudate* extends to peritoneal surface.

- Inflammation of the wall of the appendix causes venous congestion, which may compromise arterial inflow, leading to *ischaemia* and *infarction*.

Pathogenesis

- ***Organisms** from the lumen of the appendix enter the submucosa through an **ischaemic ulcer**, causing liquefaction of the wall and ultimately **perforation**.*
- As a result of the transmural inflammation, small bowel and omentum adhere to the appendix, creating a localized area of sepsis.
- *If left untreated, this may progress to form an **appendix mass** or even an **abscess**.*
- **If perforation** occurs early in the clinical course, the inflamed area will not have had time to be walled off, and generalized peritonitis follows.



Clinical Features

History

ONSET:

- The onset of acute appendicitis is associated with the gradual onset of poorly localized **central** abdominal pain.
- After a variable amount of time, the pain moves **to the right iliac fossa** and changes in character, to become sharper, constant and well localized.

AGGRAVATING FACTORS:

It is aggravated by *movement* and *coughing*.

As described earlier, this change in the nature of the pain occurs when the parietal peritoneum overlying the appendix becomes involved in the inflammatory process.

Associated symptoms& signs:

- Symptoms (*anorexia, nausea, vomiting, diarrhoea*) and signs (*fever, fetor, pallor, abdominal distension*).
- The finding of tenderness and guarding in the right iliac fossa usually makes the diagnosis without the need for other investigations.

Examination

The temperature is usually only mildly elevated (37.3–38.5°C) and there is often a **tachycardia**.

Classically, the area of **maximal tenderness** is **over McBurney's point**, with **guarding** and **rebound (percussion) tenderness**.

Rovsing's sign: Palpation in the left iliac fossa may reproduce the pain in the right iliac fossa

Psoas stretch sign: The patient may find it painful to extend the right hip owing to irritation of the psoas muscle

DIFFERENTIAL DIAGNOSIS

1-UTI:

- An inflamed retrocaecal appendix may produce poorly localized abdominal pain, and an inflamed pelvic appendix lying close to the bladder may produce symptoms of frequency and dysuria.

2- Gastroenteritis :

An inflamed pelvic appendix lying near the rectum causes irritation and diarrhoea, and is commonly mistaken for gastroenteritis.

However, gastroenteritis almost never causes abdominal tenderness.

IMPORTANT

Differential diagnosis of acute appendicitis

- Mesenteric adenitis
- Meckel's diverticulitis
- Regional ileitis (Crohn's disease)
- Carcinoma of the caecum

Gynaecological disorders

- Ruptured ovarian follicle (Mittelschmerz)
- Acute salpingitis
- Ruptured ectopic pregnancy
- Torsion of an ovarian cyst

Genitourinary

- Pyelonephritis
- Ureteric colic
- Urinary tract infection
- Right-sided testicular torsion

Special Patient Groups

Acute appendicitis is **most dangerous** in the very young, the very old and pregnant women.

As it is **uncommon** under the age of 2 years, when it does occur, it is often incorrectly diagnosed as gastroenteritis.

The symptoms and signs are **atypical** and generalized peritonitis quickly develops.

In contrast, in elderly patients, the onset is more insidious.

In the pregnant patient, the appendix is displaced upwards by the enlarged uterus, and the site of the pain and tenderness is high in the abdomen.

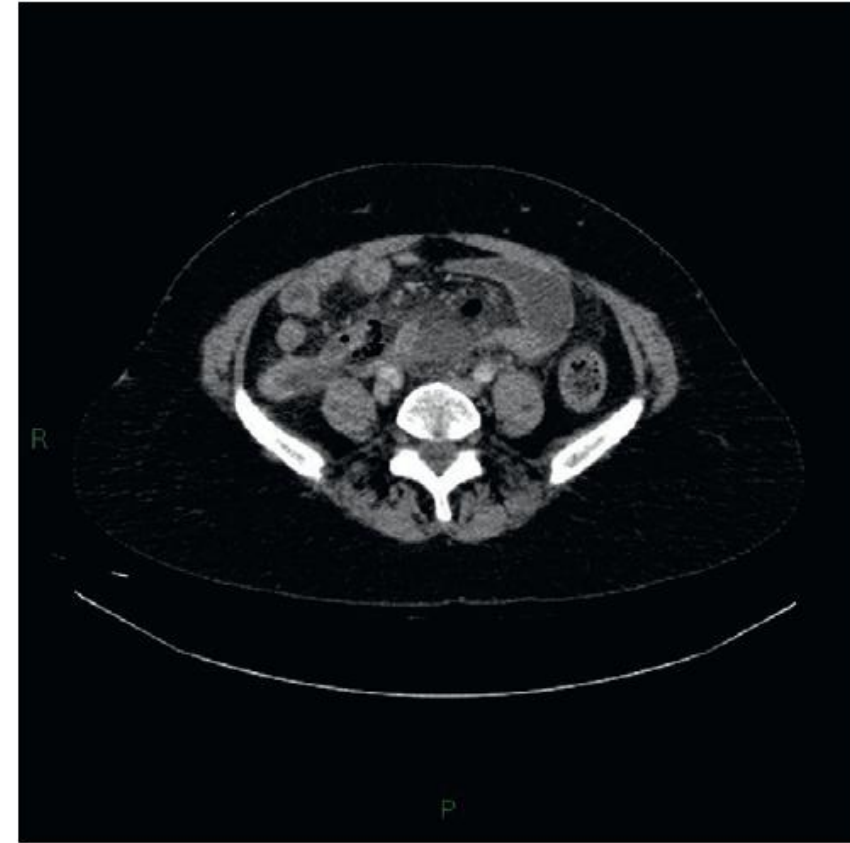
Appendicitis in pregnancy carries a high rate of morbidity and mortality for both mother and fetus.

Complications

- **Gangrenous appendicitis and perforation** tend to occur after a significantly more prolonged period of pain than uncomplicated appendicitis.
- **Generalized peritonitis** results if the inflamed area is not walled off by omentum and loops of bowel.
- **Appendix mass or an abscess** will develop if walling off occurs.
- **A perforated pelvic appendix** will lead to a pelvic abscess, and on examination there may be very little in the way of abdominal signs.

Investigations

- The diagnosis of acute appendicitis is based on **clinical assessment** and there are no specific diagnostic tests.
- **Ultrasonography** demonstrates a swollen non-compressible appendix, free fluid or even a mass in the right iliac fossa.
- *If, after clinical assessment, the diagnosis remains in doubt, the clinician must proceed along one of two lines:*
 - Either to carry out **laparoscopy** and undertake appendicectomy if indicated,
 - Or to institute a short policy of close and repeated **observation** with reassessment every hour.
- On occasions when there is diagnostic difficulty, and particularly in the elderly, **abdominal CT** can be helpful in making the diagnosis in addition to excluding other conditions.



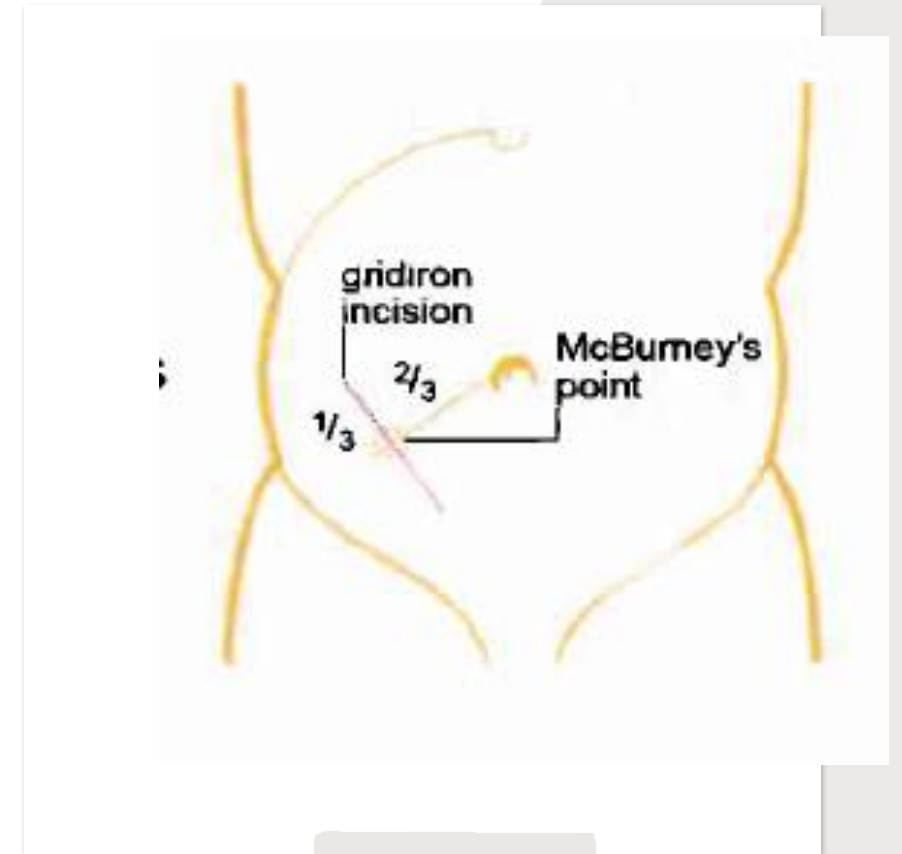
CT scan demonstrating perforated appendicitis. The thickened and inflamed appendix can be seen extending from the caecum medially associated with some flecks of gas near the tip.

Management

- The treatment of appendicitis is almost always surgical; laparoscopic appendicectomy is being carried out, especially if a diagnostic laparoscopy has been performed first to establish the diagnosis.
- It is also possible to treat patients who do not have overt peritonitis non-operatively with antibiotics.

Appendicular mass& Abscess:

- If, by the time the patient presents, a mass can be felt, non-operative management with IV fluids and antibiotics is the treatment of choice, provided there are no signs of peritonitis (when an operation should be carried out).
- In these patients, *an ultrasound* scan should be arranged to look for an underlying abscess; if one is present, it should be *drained either under radiological guidance or surgically*.



Appendicular mass& Abscess: (*Continued*)

- Following successful non-operative treatment of an appendix mass, it used to be traditional practice to carry out **an interval appendicectomy** 3–6 months later.
- This prevents further attacks, and in the elderly makes sure that there is no underlying *carcinoma of the caecum*.
- It is important, especially in older patients, to exclude a carcinoma of the caecum by either double-contrast barium enema or colonoscopy.

INTESTINAL OBSTRUCTION

Definition:

Arrest of downward propulsion of the intestinal contents

Classification:

A- ACCORDING TO PATHOLOGY:

1-SIMPLE OBSTRUCTION: due to mechanical occlusion of the gut lumen without interference of blood supply.

2- Strangulation: Interference of blood supply

3- Paralytic ileus: due to loss of propulsive power of the gut leading to functional obstruction

B- ACCORDING TO THE LEVEL OF OBSTRUCTION:

1-High small bowel obstruction

2- Low small bowel obstruction

3-Large bowel obstruction

C- ACCORDING TO ONSET& COURSE OF OBSTRUCTION:

1-Acute obstruction: intussusception

2- Chronic obstruction: cancer colon

D- ACCORDING TO MOTILITY:

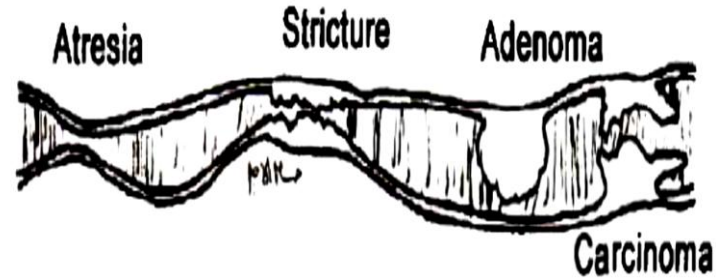
1- Dynamic obstruction: mechanical occlusion

2- Adynamic obstruction: paralytic ileus

Acute Intestinal Obstruction (according to pathology)

Causes:

1- Simple:



In the lumen (intraluminal) e.g. faecal impaction, gall stone ileus and parasitic infestation.

In the wall (intramural) e.g. Tumors, chronic diverticulitis, Crohn's disease.

Outside the wall (extramural) e.g. adhesions, strangulated hernia and volvulus.

2- Strangulation:

Strangulated hernia, volvulus, intussusception & MVO.

Clinical Picture of Simple IO

Symptoms {Pain, Distention, Vomiting, Absolute constipation[☑]}

1- **Pain:** (due to hyperactive peristalsis) usually the 1st symptom

- Onset: -in high small IO → within a day
 -In low small IO → within 2-3 days
- Character: colicky.
- Site:
 - In high small IO → above the umbilicus.
 - In low small IO → around and below the umbilicus.
 - In large IO → lower & peripheral parts of the abdomen.

- Painless → if paralytic ileus.
- Severe stabbing pain → in case of ischemia.

2- **Distension:**

3- Vomiting: By anti-peristaltic movement

- **Content:**
 - First: gastric juice (whitish mucoid).
 - Then: bile stained (greenish).
 - Finally: feculent (brown and offensive).

Vomiting may be absent in paralytic ileus[☑].

4- Absolute Constipation:

- **Definition:** No faeces or flatus is passed *inspite of desire*.

SIGNS

General	Local			
Dehydration: Dry tongue, Tachycardia, Hypotension, Oliguria	Inspection	Palpation	Percussion	Auscultation
	1-Abdominal Distension	For any mass	Tympanitic abdomen	For intestinal sounds; Exaggerated in simple obstruction or absent(paralytic ileus)
	2-Visible Peristalsis			
	3-Evidence of the cause; obstructed hernia, scar of previous operation			

Examination (continued)

- DRE (normally empty):
 - Red current jelly stool or head of intussusception.
 - Carcinoma of the rectum
 - Impacted feces in the rectum (elderly bed-ridden pt).

When to suspect strangulation?

➤ Symptoms:

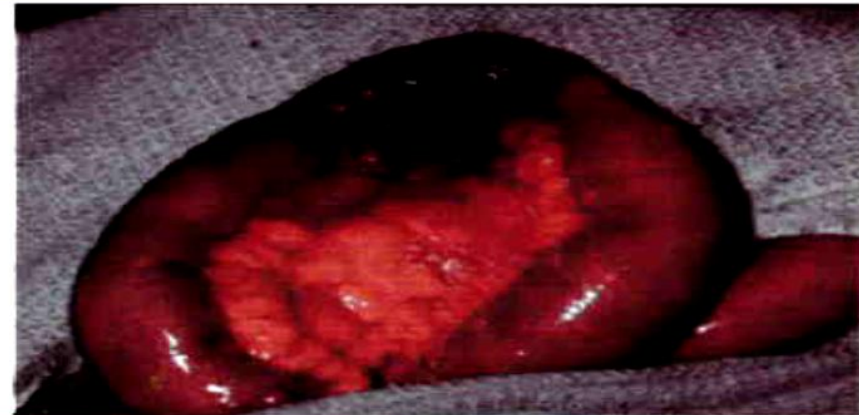
- Pain is not relieved by NG suction (as it is caused early by hyperperistalsis and ischemia, so decompression does not relieve ischemic pain)

➤ Signs:

- General: Toxic patient, fever or shocked according to the length of the segment.
- Local: Rigidity, marked tenderness and rebound tenderness.

➤ Investigations:

- CBC: leucocytosis.



This loop of intestine was strangulated within a hernia.

INVESTIGATIONS

- **Plain X-Ray of the abdomen:**
 - Erect: multiple fluid levels[□]. → High SIO → minimal fluid levels.
→ Low SIO → multiple central fluid levels.
 - Supine:
 - Jejunal loop → mucosal folds (Valvulae conniventes) crossing from one side of the lumen to the other & regular interval.
 - Ileal loop → featureless with no mucosal pattern.
 - Colon → haustrations do not reach the other side of the lumen.
- **U/S:** may show:
 - a. Distended bowel loops.
 - b. A mass of intussusception can also be demonstrated.
- **Barium enema or gastrografin enema**

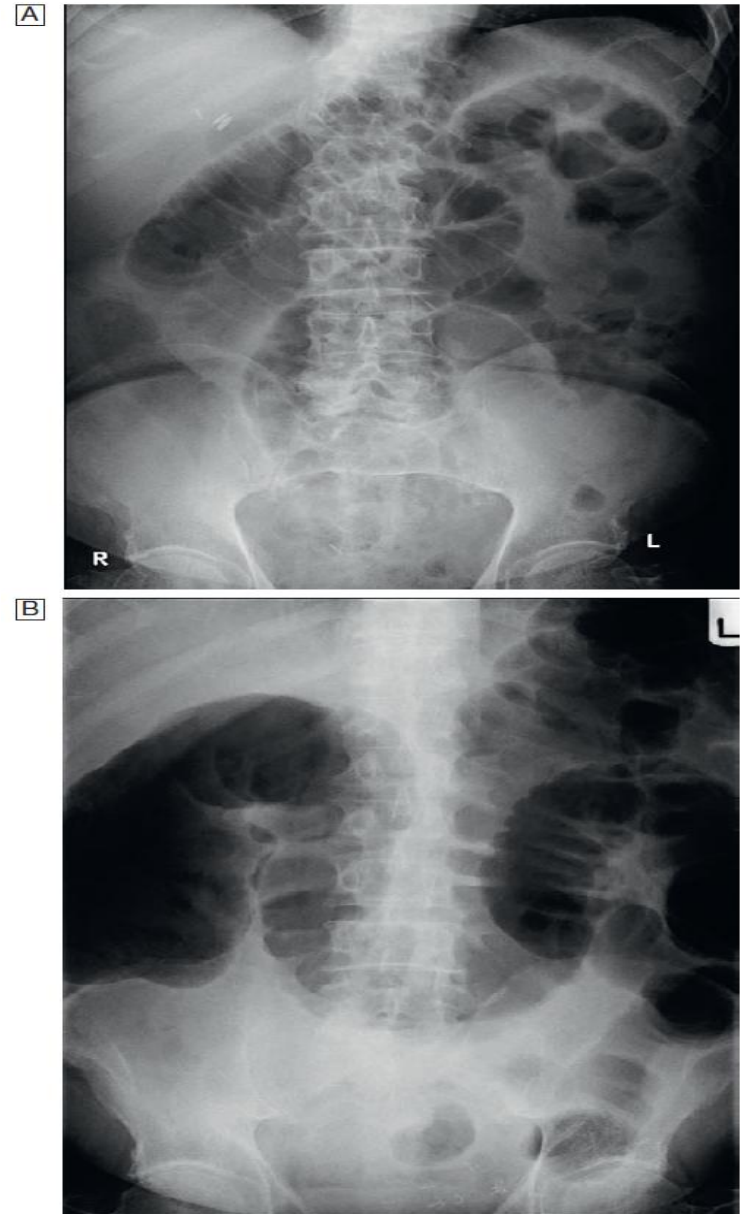
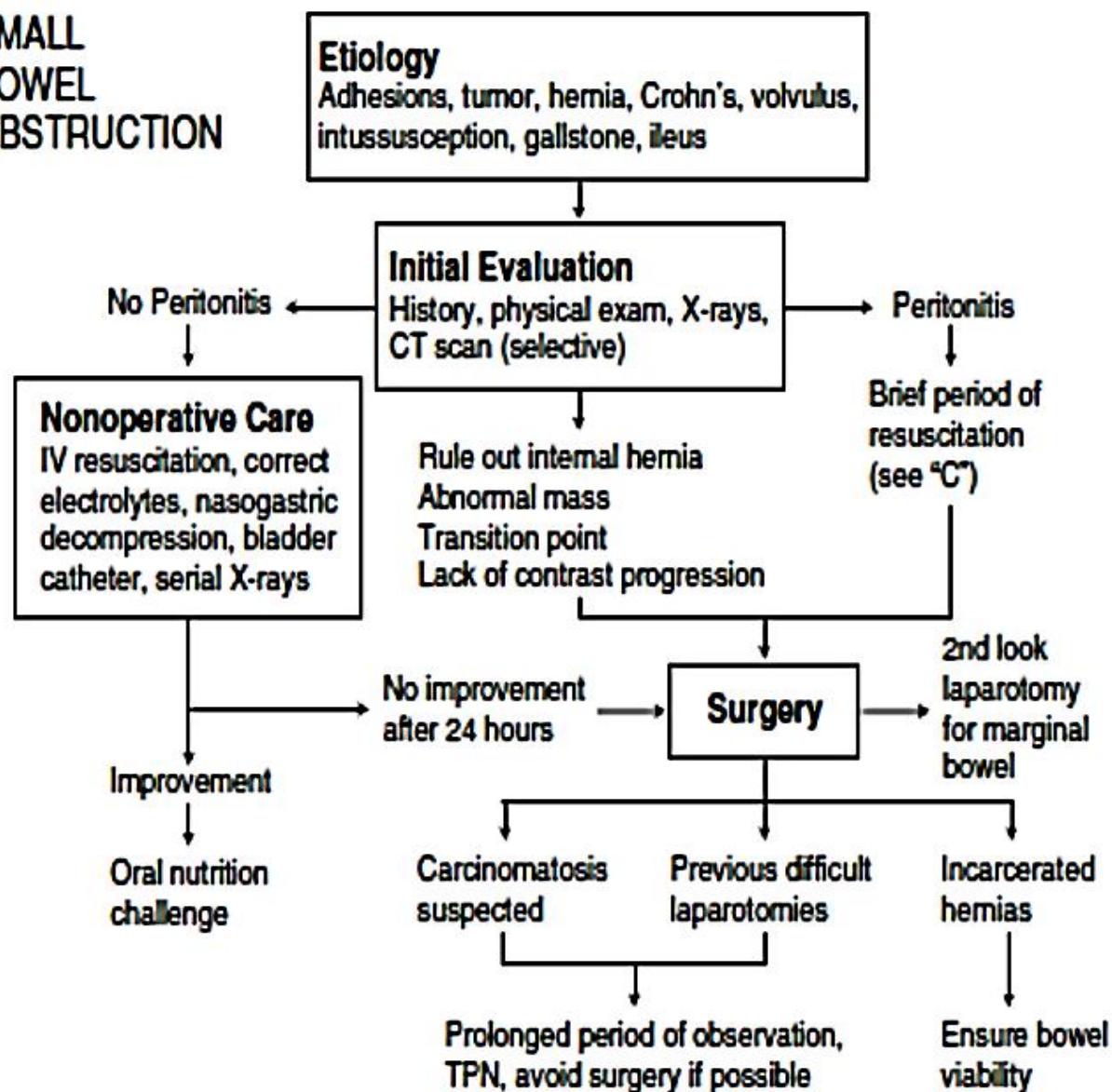


Fig. 16.15 Plain abdominal X-rays showing obstruction. [A] Small bowel obstruction. [B] Large bowel obstruction.

Algorithm for treatment of small bowel obstructions. *CT* computed tomography, *IV* intravenous, *TPN* total parenteral nutrition

SMALL BOWEL OBSTRUCTION



Special ttt for causes of Small IO

- **Adhesive small bowel obstructions** from numerous abdominal operations are frequently technically difficult to manage due to loss of tissue planes and distorted anatomy.
- Unless a patient has clearly failed medical treatment, surgery should be deferred until after a long nonoperative course of NPO and total parenteral nutrition (TPN).
- **Hernias that are incarcerated**, but not strangulated, may be reduced.
- These patients require close monitoring for evidence of bowel ischemia.
- Conversely, patients with **strangulated hernias** causing SBO may require bowel resection.

Management of Right-Sided Large Bowel Obstruction



A right hemicolectomy and primary anastomosis can be safely performed in most cases unless perforation with a significant amount of contamination has occurred.



In this instance, resection with end ileostomy and mucous fistula is preferred.



If a cancer is bulky and unresectable, intestinal bypass may be considered as an alternative.

Management of Lt-Sided Large Bowel Obstruction

Obstruction of the left colon presents a more challenging management decision, since a large column of feces exists above the point of obstruction.

It is dealt with by a variety of surgical procedures that include:

- Resection and reanastomosis with or without a stoma***
- Colostomy and Resection (Hartmann's procedure)***
- Colostomy without resection.***

Thank



You